

4.0. FUNCTIONAL DESCRIPTION

4.1. System Address Map

A Pentium Pro system based on the 440FX PCIs set supports 4 Gbytes of addressable memory space and 64 Kbytes of addressable I/O space. The lower 1 Mbyte of this addressable memory is divided into regions which can be individually controlled with programmable attributes such as disable, read/write, write only, or read only (see Register Description section for details on attribute programming).

NOTE

The Pentium Pro processor family can have up to 64 Gbytes of addressable memory. The PMC claims any access over 4 Gbytes by terminating the transaction (without forwarding it to the PCI bus). Writes are terminated by dropping the data and the PMC returns all zeros for reads

4.1.1. MEMORY ADDRESS RANGES

Figure 3 represents system memory address map. It shows the main memory regions defined and supported by the 440FX PCIs set. At the highest level, the address space is divided into four conceptual regions (Figure 3). These are the 0–1-Mbyte DOS Compatibility Area, the 1-Mbyte to 16-Mbyte Extended Memory region used by ISA, the 16-Mbyte to 4-Gbyte Extended Memory region, and the 4-Gbyte to 64-Gbyte Extended Memory region introduced by 36 bit addressing.

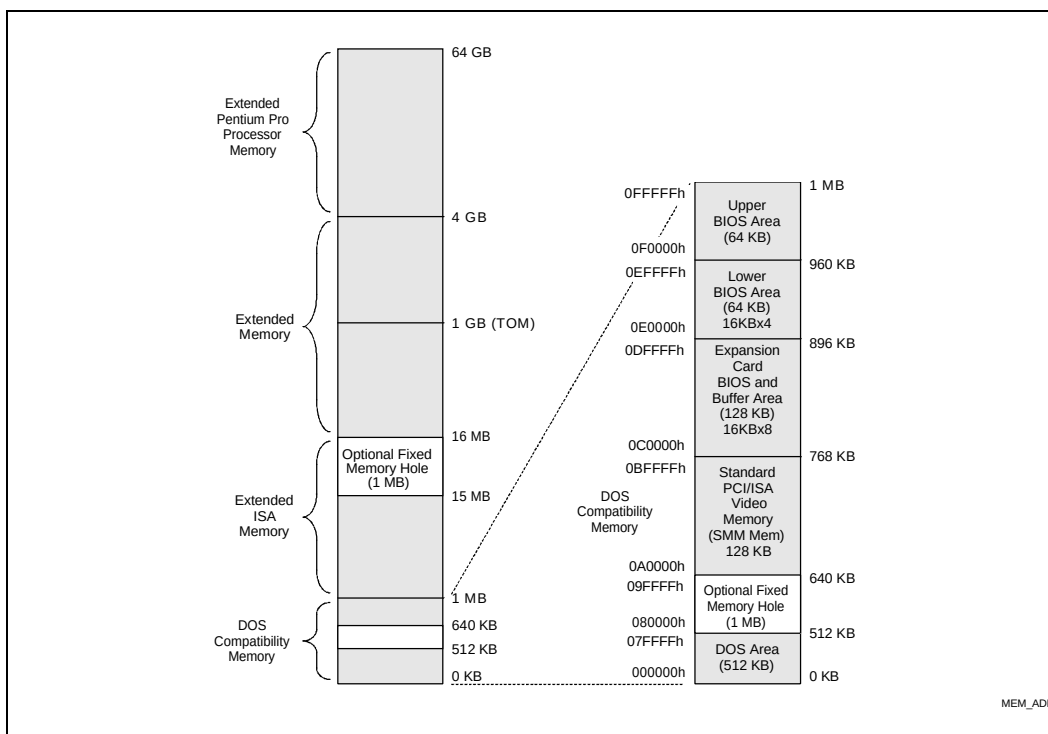


Figure 3. Memory Address Map